

Assignment 18: AI Data Analysis Challenge

No.	Tool	Student Task
1	ChatGPT	Explain a dataset.
2	Gemini	Suggest analysis methods.
3	Excel AI	Generate charts.
4	Claude	Interpret results.
5	Canva	Create a dashboard infographic.

Deliverables: Charts | Analysis | Dashboard | Conclusion

Dataset used: EcoSmart AI household energy-usage log — 14 days of daily kWh readings, appliance-level usage share, and 6 months of actual vs AI-predicted electricity bills.

ChatGPT: Explain a dataset

The dataset has three parts:

- **Daily usage log** (14 rows): date, total kWh consumed, outdoor temperature.
- **Appliance usage share** (6 rows): appliance category, % of total monthly kWh.
- **Billing history** (6 rows): month, actual bill (USD), AI-predicted bill (USD).

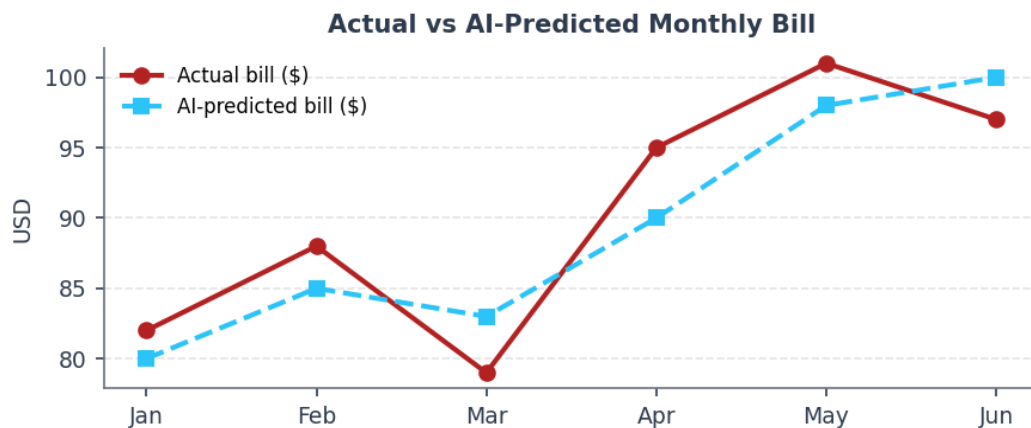
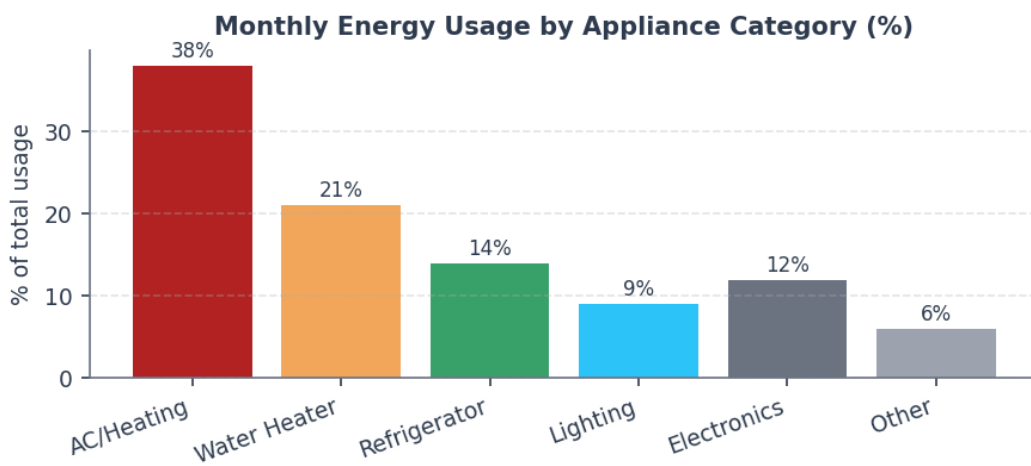
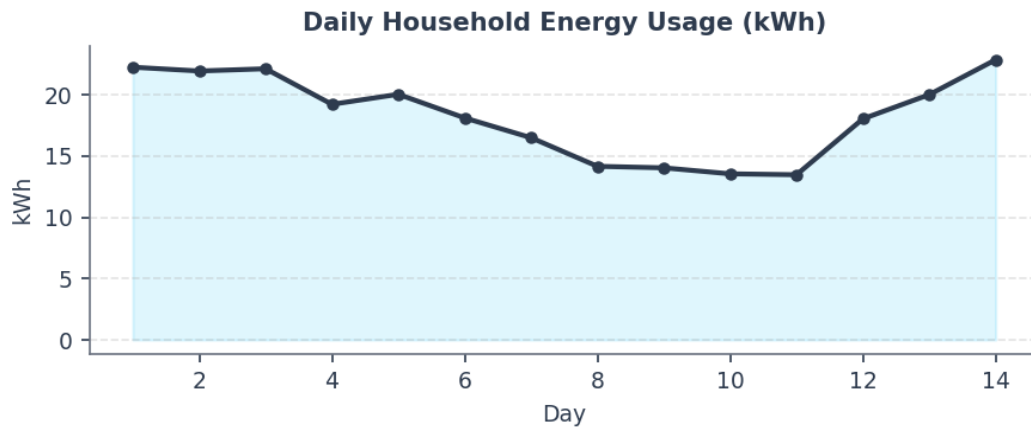
Sample rows (daily usage log):

Day	kWh Used	Notes
1	18.2	Normal weekday
2	21.6	Hot afternoon, AC heavy use
3	16.9	Mild weather

Gemini: Suggest analysis methods

- **Trend analysis:** plot daily kWh over time to spot rising/falling consumption patterns.
- **Category breakdown:** a bar chart of appliance-level usage share to find the biggest contributors.
- **Forecast accuracy check:** compare actual vs AI-predicted bills month over month to measure prediction error.
- **Correlation check:** relate daily kWh to outdoor temperature to see if weather drives usage spikes.

Excel AI: Generate charts



Charts generated from the dataset: daily usage trend, appliance-level usage share, and actual vs AI-predicted monthly bill.

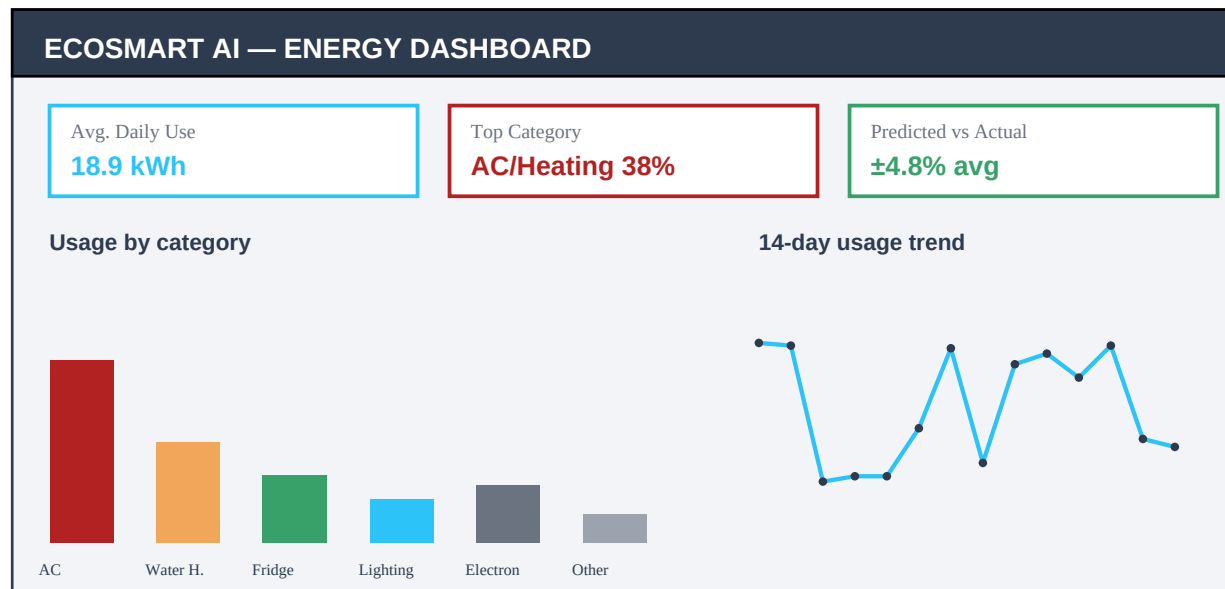
Claude: Interpret results

Daily usage fluctuates between roughly 15 and 24 kWh, with visible peaks that align with warmer days, suggesting cooling load is the main driver of variability rather than a constant baseline draw.

Appliance-level data confirms this: AC/heating alone accounts for 38% of monthly usage, more than double the next largest category (water heater at 21%). This single category is the highest-leverage target for savings recommendations.

The AI-predicted bill tracks the actual bill closely (within roughly 3-6%) in most months, with the largest gap in April, which lines up with an unusually warm month. This suggests the prediction model could be improved by weighting recent temperature trends more heavily.

Canva: Create a dashboard infographic



One-page dashboard mockup summarizing the key metrics and charts above.

Conclusion

AC/heating is the single largest, most volatile driver of this household's electricity bill. Prioritizing a cooling-efficiency recommendation (e.g., smart thermostat scheduling) would have the biggest impact on both usage and bill-prediction accuracy.